

Arrival choices

You can begin without previous learning. Word help: tectonic means related to movement of Earth's plates.

Start with 1 and 2. Try 3 and 4 when ready. Reading and exact scribing are available.

1 What does tectonic mean?

Support: Use the reminder strip or talk it through with an adult.

My response

2 Which meaning fits tectonic?

Support: Choose: Related to movement of Earth's plates OR People or things in harm's way.

My response

3 What caused the 2011 tsunami?

Support: Use the model evidence anchor A or read one relevant card together.

My response

4 Is magnitude the same as risk?

Support: Give a reason when ready.

My response

Stretch: use a precise example or explain which detail supports your answer.

Evidence cards

Read, point or listen. Keep the source label with any detail you use.

A Japan 11 March 2011

A magnitude 9.1 earthquake occurred off north-east Honshu, Japan. Sudden movement on the subduction boundary displaced the seafloor and produced a tsunami.

Source: USGS event record, official20110311054624120_30

Limit: Magnitude measures the earthquake; it is not a direct count of people affected.

B Effects and responses

Shaking and coastal flooding damaged places and disrupted people's lives. Earthquake-resistant design addresses shaking; warnings, evacuation routes and higher ground address tsunami exposure.

Source: Authored explanation of the case and risk-management principles

Limit: A sea wall or warning cannot eliminate all risk. Do not judge people without considering access and warning time.

C Heat and unequal risk

On 19 July 2022 the Met Office recorded 40.3°C at Coningsby, 40.2°C at St James's Park and 39.6°C at Topcliffe. Outdoor work increases exposure; difficulty accessing cool spaces can increase vulnerability. Shade and accessible cool spaces can reduce harm.

Source: Met Office July 2022 report, station table; risk explanation authored for teaching

Limit: This explains risk factors, not a medical assessment of an individual.

D Reading a hazard feed

A USGS event row can include UTC time, place, magnitude and depth in kilometres. A short feed is a dated snapshot. It cannot predict the next earthquake.

Source: USGS earthquake feed documentation

Limit: Small samples, reporting thresholds and later revisions affect patterns.

Visual resource

Use this visual with the evidence cards and enquiry questions.



Source: Natural Earth public-domain outlines, labels simplified. Terms of use:
<https://www.naturalearthdata.com/about/terms-of-use/> (recorded in Sources and checks).

A detail I noticed and what it helps me explain

Shared investigation

1. Locate the Japan case and model subduction, sudden movement and seafloor displacement.
2. Interpret the event's magnitude and map.

Work with a partner or adult, or use your own table. One person suggests; the other checks the evidence. Swap after one turn. Passing is allowed.

Move or match the cards

Cards to use
Warnings and accessible evacuation
Sudden movement and shaking
Plate motion and strain

Possible labels: Cause / Event / Response

Our explanation

Support: Use the glossary and one chunk of evidence. Plan the claim and supporting detail before explaining.

Further challenge: Explain why equal-magnitude earthquakes can have different consequences.

Supported independent task

Complete this route only. Explain causes of tectonic hazards.

1. Build an evidence-based cause, effects and responses diagram.
2. Interpret a map or graph using one accurate value.
3. Make a case-study judgement distinguishing hazard from risk.

Support: Use the glossary and one chunk of evidence. Plan the claim and supporting detail before explaining. Complete the organiser one space at a time. At the boundary _____. Sudden movement _____. This caused _____. Risk also depends on _____.

Cause and supporting detail

Event or change

Consequence and supporting detail

How the links connect

An adult may read or scribe your exact response. They should not choose the answer.

Standard independent task

Complete this route only. Explain causes of tectonic hazards.

1. Build an evidence-based cause, effects and responses diagram.
2. Interpret a map or graph using one accurate value.
3. Make a case-study judgement distinguishing hazard from risk.

Support: At the boundary _____. Sudden movement _____. This caused _____. Risk also depends on _____.

Cause and supporting detail

Event or change

Consequence and supporting detail

How the links connect

Check the required evidence and revise one detail.

Stretch independent task

Complete this route only. Explain causes of tectonic hazards.

1. Build an evidence-based cause, effects and responses diagram.
2. Interpret a map or graph using one accurate value.
3. Make a case-study judgement distinguishing hazard from risk.
4. Explain why equal-magnitude earthquakes can have different consequences.

Support: Choose the evidence independently. Use the frame only if it helps.

Cause and supporting detail

Event or change

Consequence and supporting detail

How the links connect

Explain why equal-magnitude earthquakes can have different consequences.

Exit ticket

Choose a response mode. You may give your answers privately to an adult.

What caused the 2011 tsunami?

My response

Is magnitude the same as risk?

My response

Support: At the boundary _____. Sudden movement _____. This caused _____. Risk also depends on _____.

Further challenge: Explain why equal-magnitude earthquakes can have different consequences.

One thing I changed or want to check
